Chapter 12. Nonparametric Regression

Kernel regression, local polynomial regression, bandwidth and order selection, confidence intervals

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Model

$Y_i = m(X_i) + \sigma(X_i)\epsilon_i$ with $E\epsilon_i = 0$, $\operatorname{Var}\epsilon_i = 1$ and $m(x) = E(Y \mid X = x)$ smooth but otherwise unrestricted. The goal is to estimate the curve m and quantify its uncertainty.

12.1 Local polynomial regression

Definition (Local polynomial estimator)

With $K_h(u) = K(u/h)/h$, fit at each x

$$\hat{\boldsymbol{\beta}}_x = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \big(Y_i - \beta_0 - \beta_1(X_i - x) - \cdots - \beta_p(X_i - x)^p\big)^2 K_h(x - X_i),$$

and set $\hat m_p(x) = \hat\beta_{0,x}$. In matrix form, $\hat{\boldsymbol{\beta}}_x = (X_x' W_x X_x)^{-1} X_x' W_x \mathbf{y}$, with $W_x = \operatorname{diag}(K_h(x - X_i))$ and X_x having rows $(1, X_i - x, \ldots, (X_i - x)^p)$.

Every local polynomial estimator is linear in the responses: $\hat m_p(x) = \sum_i w_i^p(x) Y_i$.

Proposition 12.1

With $S_q = \sum_j K_h(x - X_j)(X_j - x)^q$:

(i) $p = 0$ (**Nadaraya–Watson**): $w_i^0(x) = K_h(x - X_i)/\sum_j K_h(x - X_j)$;

(ii) $p = 1$ (**local linear**): $w_i^1(x) = \dfrac{\{S_2 - (X_i - x)S_1\}K_h(x - X_i)}{S_0 S_2 - S_1^2}$.

In both cases $\sum_i w_i(x) = 1$. For $p = 1$ also $\sum_i w_i^1(x)(X_i - x) = 0$: the fit reproduces straight lines exactly.

Proof. (i) Minimize $\sum (Y_i - \beta_0)^2 K_h(x - X_i)$. (ii) $X_x' W_x X_x = \begin{pmatrix} S_0 & S_1 \\ S_1 & S_2 \end{pmatrix}$, and the first row of its inverse is $(S_2, -S_1)/(S_0 S_2 - S_1^2)$. Multiplying by $X_x' W_x$ gives the weights. Summing, $\sum_i (S_2 - (X_i - x)S_1)K_h = S_2 S_0 - S_1^2$, and $\sum_i (S_2 - (X_i - x)S_1)(X_i - x)K_h = S_2 S_1 - S_1 S_2 = 0$. ∎

As $h \to \infty$ the Nadaraya–Watson estimator tends to the constant $\bar Y$, while the local linear estimator tends to the least squares line.

Theorem 12.2

Let m be twice continuously differentiable, let the design density f be smooth with $f(x) > 0$, and use the Gaussian kernel. As $n \to \infty$, $h \to 0$ and $nh \to \infty$, conditionally on the X's and for $p = 0, 1$,

$$\operatorname{bias}\hat m_p(x) = B_p(x)h^2 + o_P(h^2), \qquad \operatorname{Var}\hat m_p(x) = \frac{\sigma^2(x)}{2\sqrt{\pi} f(x)} \cdot \frac{1}{nh} + o_P\Big(\frac{1}{nh}\Big),$$

with $B_0 = \dfrac{m''}{2} + \dfrac{m' f'}{f}$ and $B_1 = \dfrac{m''}{2}$.

Sketch. Since the weights sum to 1, bias $= \sum_i w_i(x)\{m(X_i) - m(x)\}$ and Var $= \sum_i w_i^2 \sigma^2(X_i)$. For $p = 0$, the numerator $\frac{1}{n}\sum K_h(x - X_i)\{m(X_i) - m(x)\}$ has mean $\int K(u)\{m(x - hu) - m(x)\} f(x - hu)\, du = h^2(\frac{m'' f}{2} + m' f') + o(h^2)$. The denominator tends to $f(x)$, giving B_0. For the variance,

$\frac{1}{n^2}\sum K_h^2\sigma^2 \approx \frac{\sigma^2 f}{nh}R(K)$ with $R(\phi) = \frac{1}{2\sqrt{\pi}}$; divide by f^2. For $p = 1$, the property $\sum w_i^1(X_i - x) = 0$ removes the first-order term $m'(x)(X_i - x)$ exactly, including the asymmetry of the design that produced $m'f'/f$. The leading variance is unchanged. ■

The local linear estimator has the same leading variance as Nadaraya–Watson but no **design bias** $m'f'/f$. It also behaves better near the boundary, which is why it is the default. Generally the bias decreases with the degree, $|\text{bias}\,\hat{m}_0| \gtrsim |\text{bias}\,\hat{m}_1| \gtrsim |\text{bias}\,\hat{m}_2| \gtrsim \dots$. The leading variance follows the pattern

$$\text{Var}\,\hat{m}_0 \approx \text{Var}\,\hat{m}_1 < \text{Var}\,\hat{m}_2 \approx \text{Var}\,\hat{m}_3 < \cdots .$$

Raising an even degree to the next odd one therefore reduces bias at no cost in variance, which is why odd degrees (usually $p = 1$, sometimes $p = 3$) are preferred. The bias $\frac{m''}{2}h^2$ is positive in valleys and negative at peaks, so the estimate flattens features of the curve.

12.2 Choosing the bandwidth

Integrating the bias and variance against the design density,

$$\text{MISE} \approx \frac{h^4}{4}\int (m'')^2 f + \frac{\int \sigma^2}{2\sqrt{\pi}\,nh}, \qquad h^* = \Big(\frac{\int \sigma^2}{2\sqrt{\pi}\int (m'')^2 f \cdot n}\Big)^{1/5}.$$

- **Direct plug-in** (Ruppert–Sheather–Wand): estimate $\int (m'')^2 f$ and $\int \sigma^2$ by blockwise quartic fits, with the number of blocks chosen by an information criterion.
- **Leave-one-out cross-validation:** minimize $\text{CV}(h) = \frac{1}{n}\sum_i \{Y_i - \hat{m}^{(-i)}(X_i)\}^2$.

Proposition 12.3

For a linear smoother whose leave-one-out fit equals the full fit at X_i after deleting point i (true for local polynomial fits), $\text{CV}(h) = \dfrac{1}{n}\sum_i \Big(\dfrac{Y_i - \hat{m}(X_i)}{1 - w_i(X_i)}\Big)^2$.

Proof. Let $\tilde{Y}_i = \hat{m}^{(-i)}(X_i)$. Replacing Y_i by $\tilde{Y}_i$ leaves the weighted fit at X_i unchanged: the added point lies exactly on the leave-one-out local fit, so it does not change the weighted least squares solution. Hence $\tilde{Y}_i = \sum_{j\neq i} w_j(X_i)Y_j + w_i(X_i)\tilde{Y}_i$, so $\tilde{Y}_i = \frac{\hat{m}(X_i) - w_i(X_i)Y_i}{1 - w_i(X_i)}$ and $Y_i - \tilde{Y}_i = \frac{Y_i - \hat{m}(X_i)}{1 - w_i(X_i)}$. ■

So CV needs only one fit per h.

12.3 Confidence intervals

As for densities, intervals target the smoothed curve $m_h(x) = E\hat{m}_1(x)$.

- **Theory-based:** $\widehat{\text{se}}(\hat{m}_1(x)) = \big(\sum_i w_i^1(x)^2\hat{\sigma}^2(X_i)\big)^{1/2}$ and $m_h(x) \in \hat{m}_1(x) \pm z_{\alpha/2}\widehat{\text{se}}$. Under homoscedasticity, with W the $n \times n$ smoother matrix ($\hat{\mathbf{y}} = W\mathbf{y}$),

$$\hat{\sigma}^2 = \frac{\|(I - W)\mathbf{y}\|^2}{\text{tr}\{(I - W)'(I - W)\}}.$$

  Under heteroscedasticity, smooth $\log r_i^2$ against X_i and exponentiate, which keeps $\hat{\sigma}^2(x) > 0$.
- **Bootstrap:** resample pairs (X_i, Y_i), refit with the *same* h on a grid, and use $\hat{m}_1(x) \pm z_{\alpha/2}\widehat{\text{se}}^*(x)$ or the percentile interval $(Q^*_{\alpha/2}(x), Q^*_{1-\alpha/2}(x))$. Residual bootstraps need care because the residuals inherit the smoothing bias.

Simultaneous bands are possible, as in Section 11.4, but are rarely used in practice.